Retuned translated probes and the conditional bound $\Lambda < 0.116398$ for the de Bruijn–Newman constant

Dale James Raymond
ORCID: 0009-0001-1682-8529
dale_raymond@icloud.com

10 October 2026 — Version 1.0

Abstract

Assume that the Riemann zeta function has no nontrivial zeros in the open half-plane $\operatorname{Re} s > 7/8$. A computer-assisted argument gives

$$\Lambda \le \frac{116397726408291}{10^{15}} = 0.116397726408291 < 0.116398$$

for the de Bruijn–Newman constant in the Polymath normalisation. The proof combines a retuned probe-height schedule, matched finite logarithmic coefficients and exact translated-kernel inequalities. The source calculation preserves common multiplicative phases and cancels shared convolution terms before bounding the remaining terms. A family of 44 rational kernel certificates then adapts the comparison to the smaller height-ratio domains along the descent. Together these estimates improve the complete contraction rate, including the entire distinguished-pair subtraction. The physical evidence covers 608 closed time cells, six frequency classes including an unbounded final class, and all eleven translated class pairs. The retuned physical enclosures were produced in two interval-arithmetic constructions. Two 14,899-step rational histories reproduce byte-for-byte and pass a separately implemented auditor. The paper gives the global first-contact argument, the corrected Gaussian remainder and the continuous-domain kernel proof; the analytic supplements, editable sources and executable evidence accompany it. The 7/8 assertion is an explicit hypothesis, not a conclusion of the numerical checks.

2020 Mathematics Subject Classification. Primary 11M26; secondary 65G30.
Keywords. de Bruijn–Newman constant; translated probes; matched logarithmic coefficients; exact polynomial certificates; interval arithmetic.

1 Introduction

The de Bruijn–Newman constant Λ is the threshold at which all zeros of the heat-deformed Riemann xi function are real. Heat-flow estimates can combine a time-zero strip, finite localisation walls and logarithmic-derivative bounds away from the zeros to give a quantitative upper bound [5, 6]. This paper strengthens the translated-probe construction of [1] under an explicit narrower initial strip.

For $1/2 < \alpha \le 1$, let H_α mean that every nontrivial zeta zero ρ satisfies $\operatorname{Re}\rho \le \alpha$. Zeros on the boundary are allowed.

Theorem 1 (Conditional upper bound)**.** *Assume* $\mathsf{H}_{7/8}$*. In the normalisation of Section 2,*

$$\boxed{\Lambda \le \frac{116397726408291}{10^{15}} = 0.116397726408291 < 0.116398.} \tag{1}$$

1.1 Optimising the complete contraction rate

A useful probe must do more than give a large logarithmic-derivative lower bound. Its height, its translation and its kernel constant also determine how much zero contraction that bound supplies. The quantity to improve is

$$c = \frac{2\lambda}{h}\left(\ell - \frac{M}{h}\right),$$

where ℓ is the complete physical source floor and M/h removes the whole contribution of the distinguished conjugate pair. Optimising only ℓ or only λ can choose the wrong probe.

The first step makes the trade-off concrete. The earlier prepared comparison used height 2.18, a source floor about 6.3340 and rate about 10.3210. The present comparison uses height 1.88, a smaller source floor about 6.0650, and a larger rate about 11.1631. The smaller source floor is more useful after the height and own-pair terms are included. These rounded values illustrate exact rational data, not sampled physical measurements.

Three changes supply the improvement. First, the kernel family certifies larger constants on smaller height-ratio domains. Second, the physical estimates are regenerated at 274 changed probe heights. Third, the matched-logarithmic calculation retains the shared finite convolution before estimating the remainder. The final schedule selects one complete certified source row in each time cell and is frozen before the exact histories are generated. No source bound is moved to an uncertified height.

Construction in the retained development history	Exact primary ceiling
Earlier undeposited comparison	0.118068908899880
New kernels; unchanged physical source	0.117794794642445
Retuned heights; basic joint source	0.116787767525899
Retuned heights; matched logarithmic source	**0.116397726408291**

The total reduction is 0.001671182491589, approximately 1.41543% of the earlier draft ceiling. All four comparisons use the same strict seed $9/16 + 10^{-6}$ and 32 substeps per source cell. The gain is therefore not from tightening the seed or increasing the time subdivision. This is a comparison of completed constructions, not a global-optimality claim.

1.2 Context, assumptions and scope

The 7/8 assertion appears in the supplied OpenAI manuscript of 30 September 2026 [2]. Its 5 October companion states the different 11/12 half-plane [3]. The pinned formal-source declaration is identified in [4]; no execution of that formalisation is claimed here. The mathematical assumption used in this paper is exactly $\mathsf{H}_{7/8}$, regardless of the external assertion's publication or review status.

Polymath's heat-flow framework and the finite-RH result of Platt and Trudgian are analytic inputs [5, 6]. The author's earlier public paper reports $\Lambda < 0.149396$ from the classical initial strip [1]. That result uses a different initial hypothesis from the one here. The subsequent preprints discussed there, including Gomila and Gordon [7, 8], and Planat's collective-contraction framework [9], provide context. No numerical certificate from those three works is imported into Theorem 1.

The companion *Further Consequences* records a separate implication from $\mathsf{H}_{34/43}$ to $\Lambda < 0.09$, its still-unproved covariance supplier, and a restricted limitation of the older source-table method. None is a premise of Theorem 1.

1.3 Reading the proof

Sections 2–4 give the initial strip, localisation and finite logarithmic source argument. Section 5 specifies the retuned source and its exact selection. Sections 6–7 prove kernel domination and global comparison. Section 8 identifies the exact endpoint and verification boundary. The two appendices give the complete kernel-constant table and the Gaussian correction. The supplied *Matched Source Supplement* expands the finite-source algebra; *Prior Analytic Foundations* gives the physical approximation and localisation framework, with its Gaussian bound replaced explicitly by Appendix B below.

2 Normalisation, initial separation and localisation

Use the normalisation of [5]:

$$\begin{aligned} H_t(z) &= \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,\mathrm{d}u, \\ \Phi(u) &= \sum_{n\ge 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}}, \\ H_0(z) &= \tfrac{1}{8}\xi((1+iz)/2), \qquad \partial_t H_t = -\partial_z^2 H_t. \end{aligned} \tag{2}$$

Here $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. The functions H_t are real and even. The de Bruijn–Newman constant is the threshold such that H_t has only real zeros exactly when $t \ge \Lambda$. De Bruijn's strip theorem, in [5, Theorem 3.2], contracts a zero strip of half-height Y at time t to half-height at most $\sqrt{\max(Y^2 - 2\tau, 0)}$ at time $t+\tau$.

Lemma 2 (Time-zero strip)**.** *Under* $\mathsf{H}_{7/8}$ *every zero of* H_0 *has* $|\operatorname{Im} z| \le 3/4$. *The initial squared comparison value used here is*

$$q_0 = \frac{562501}{1000000} > \frac{9}{16}, \qquad q_0 - \frac{9}{16} = \frac{1}{1000000}. \tag{3}$$

Proof. The functional equation reflects a nontrivial zero ρ to $1-\rho$, so $1/8 \le \operatorname{Re}\rho \le 7/8$. Its corresponding H_0 height in (2) is $1 - 2\operatorname{Re}\rho$. The strict seed inequality is exact. □

Ordinary contraction from this strip alone gives $\Lambda \le 9/32$. The physical estimates below supply the additional contraction needed for the stated bound.

Fix

$$X_0 = 6000000185827, \qquad N_0 = 690988.$$

The function B_t is the nonvanishing analytic normaliser defined in the included Foundations, Part A, Section 4, with the same logarithm branches and normalisation throughout. The finite-head wall is valid for $x = X_0$, $0 \le t \le 41/250$ and $0 \le y \le 1$. With the corrected physical error allowance in Appendix B, the unchanged ideal-prefix certificates imply

$$\left|\frac{H_t(X_0+iy)}{B_t(X_0+iy)}\right| > \frac{79}{100} - \frac{1}{100} - \frac{1}{180} = \frac{697}{900} > \frac{3}{4}. \tag{4}$$

Both complete wall constructions are included. The finite-RH theorem of Platt–Trudgian verifies the exact ordinate 3000175332800, which exceeds $(X_0+1)/2$ by 175239886 [6, Theorem 1]. Its original zero computation is cited, not repeated.

Initially every zero inside the finite head is real. The wall, positivity of $H_t(iy)$, the enclosing classical strip, and local forward Hermite splitting at a real multiple zero prevent entry or birth of a nonreal zero in that head during the declared time range. Thus every prospective nonreal contact is at $|\operatorname{Re} z| > X_0$. These are the localisation arguments of the included Foundations, Part A, Sections 2–3, with the safe wall (4). No later narrow-strip or finite-canopy result is imported.

3 Uniform physical source bounds at all frequencies

Define, only after proving probe nonvanishing,

$$S(x,h,t) = -\operatorname{Im}\frac{H_t'(x+ih)}{H_t(x+ih)}.$$

The selected source in `evidence/SOURCE_SELECTED.json` prescribes rational heights h_i on the 608 closed cells

$$I_i = [i/4000, (i+1)/4000], \qquad 0 \le i < 608.$$

Its heights lie in $[69/100, 47/25]$. The schedule is part of the theorem's finite data. Each comparison step uses the lower bound certified at its prescribed height, without interpolation between heights. For $s = \frac{1}{2}\log(x/(4\pi)) - \log N_0$, the six frequency classes are

$$[0,1/20],\ [1/20,3/20],\ [3/20,2/5],\ [2/5,1],\ [1,2],\ [2,\infty).$$

They cover every actual $x \ge X_0 - 2$. The source activation bounds include all native cutoffs, including future cutoffs in the unbounded final class. The translated points $x \pm a$, with $x \ge X_0$ and $|a| < 2$, belong to one of the eleven class pairs (j,j) or $(j,j+1)$. The original source proof establishes this same-or-adjacent-class restriction and handles the joins between classes.

3.1 Physical approximation and the labelled model

Put $L = \log(x/(4\pi))$, $b_t(n) = \exp(t\log^2 n/4)$ and

$$s_U^0 = (1+h)/2 + tL/4 - i(x/2 + \pi t/8), \qquad E_t(s) = \prod_{p\in\{2,3,5\}} (1 - b_t(p)p^{-s}).$$

Multiplying the *finite* first and reflected branches by E_t gives the complete labelled model

$$Q = 1 + \sum_{j>0} a_j z_j, \qquad V = \sum_{j>0} \lambda_j a_j z_j, \qquad Q_h = -V/2,$$

with $a_j, \lambda_j \ge 0$, $|z_j| = 1$ and all reflected and product-tail labels retained. Certified masses satisfy

$$\bar{A} \ge \sum a_j, \quad \bar{B} \ge \sum \lambda_j a_j, \quad \bar{A} < 4/5.$$

The source first proves nonvanishing, then divides to obtain

$$S > \ell + \frac{C - \operatorname{Re}(V/Q)}{2} + (E_\ell - \operatorname{Re}(E_h/E)). \tag{5}$$

The floor ℓ already includes subtraction of the physical errors. The term E_ℓ is the source's phase-free upper cap for $\operatorname{Re}(E_h/E)$; their difference is a nonnegative phase deficit.

The all-real-abscissa Riemann–Siegel/Gaussian derivation and its frozen-native- cutoff Cauchy disk are in the Foundations, Part A, Sections 4–5. The actual disks obey $x \ge X_0 - 4$, $0.54 \le y \le 2.41$, and $0 \le t \le 19/125$. The calculation includes the value error, the complete edge pair and the derivative error, and proves denominator exclusion. Appendix B corrects the coarse Gaussian comparison on this domain and checks the complete physical-error rows of the original and retuned source constructions. The argument at $h > 1$ uses the enlarged-height derivation rather than the restricted-height basic approximation theorem. Cauchy's estimate is applied to the holomorphic error on the frozen-cutoff disk, not to a nonholomorphic leading-model error.

3.2 Retaining the common phase

For $c_\pm \ge c_0 \ge 0$, $|z| = |u| = 1$ and $\varepsilon \in \{1, -1\}$,

$$\frac{c_+(1 - \varepsilon \operatorname{Re}(zu)) + c_-(1 - \varepsilon \operatorname{Re}(z\bar{u}))}{2} \ge c_0(1 - |\operatorname{Re} u|). \tag{6}$$

Each factor $1 - \varepsilon \operatorname{Re}(zu^{\pm 1})$ is nonnegative, so replacing $c_\pm$ by c_0 gives a lower bound. The identity $\operatorname{Re}(zu) + \operatorname{Re}(z\bar{u}) = 2 \operatorname{Re} z \operatorname{Re} u$ completes the estimate. The original translated-probe refinement applies this only to identified nonnegative cone-surplus terms and retains separate denominators at the two points. It handles the constant/head pairs through index 1024 and the first eight harmonics for the three finite Euler factors. Only the selected positive deficits contribute to the estimate; the remaining deficits are discarded throughout the calculation.

4 Matched finite logarithmic overlap

Matching the common finite-head terms gives an additional lower bound for the paired source. The matched-source supplement, Sections 3–8, contains the complete coefficient, support and numerical constructions. All logarithmic identities below are formal coefficient identities; no convergent positive-time Euler product is assumed.

4.1 A formal positivity statement

All algebra here is formal Dirichlet convolution: $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$. The function δ is its unit, δ_p is one at p and zero elsewhere, and $\log_* f$ is the formal convolution logarithm of a series with $f(1) = 1$. Write

$$\mathcal{D}f(n) = (\log n)f(n), \qquad c_t = b_t * \prod_{p \in \{2,3,5\}} (\delta - b_t(p)\delta_p), \qquad \widetilde{d}_t = \mathcal{D} \log_* c_t.$$

Lemma 3. *For every integer $n \ge 1$ and $t \ge 0$, the coefficients $c_t(n)$ and $\widetilde{d}_t(n)$ and all their time derivatives are nonnegative. Moreover $c_t(1) = 1$ and $c_t(p) = \widetilde{d}_t(p) = 0$ for $p = 2, 3, 5$.*

Proof. For $d_t = \mathcal{D} \log_* b_t$, the derivation rule and $\partial_t b_t = \mathcal{D}^2 b_t/4$ give

$$\partial_t d_t(n) = \frac{\log^2 n}{4} d_t(n) + \frac{\log n}{4} \sum_{ab=n,\ a,b\ge 2} d_t(a)d_t(b).$$

The initial coefficients are the nonnegative von Mangoldt coefficients. Induction on n and the integrating-factor formula prove positivity of all derivatives. At a selected prime power, put $\kappa = \log^2 p/4$ and $e_k = d_t(p^k) - (\log p)e^{k\kappa t}$. Then

$$e_k' = k^2 \kappa e_k + (k^2 - k)\kappa (\log p) e^{k\kappa t} + \frac{k \log p}{4} \sum_{j=1}^{k-1} d_t(p^j) d_t(p^{k-j}), \quad e_k(0) = 0.$$

Its forcing has all derivatives nonnegative. Elsewhere $\widetilde{d} = d$. Finally the exact recurrence

$$(\log n) c_t(n) = \sum_{d|n,\ d\ge 2} \widetilde{d}_t(d) c_t(n/d) \tag{7}$$

proves the assertion for c by induction. The selected primes vanish identically. □

For positive t, the infinite numerical heat-weighted Dirichlet series converges at no fixed abscissa. The argument uses the preceding identities coefficientwise, and all products evaluated below are finite.

4.2 Cancelling shared terms before enclosure

Let $\sigma = \operatorname{Re} s_U^0$ and $K = 8192$. In the complete first head through N_0, the physical phases obey $z_d z_m = z_{dm}$. Define

$$T = \sum_{2\le d\le K} \widetilde{d}_t(d) d^{-\sigma} z_d, \qquad D_K = \sum_{2\le d\le K} \widetilde{d}_t(d) d^{-\sigma},$$
$$Z = \sum_{2\le d\le K} \widetilde{d}_t(d) d^{-\sigma} \sum_{2\le m\le N_0/d} c_t(m) m^{-\sigma}.$$

The matching uses only the first-head labels, where the multiplicative relations hold. The reflected and product-tail labels remain separate. By (7), the shared mass between QT and V through the common head is $D_K + Z$. Remove it from both sums before using the triangle inequality:

$$|QT - V| \le \bar{B} + (1 + \bar{A}) D_K - 2(D_K + Z) = \bar{B} - (1 - \bar{A}) D_K - 2Z. \tag{8}$$

The right side bounds the mass of the unmatched labelled terms and is therefore nonnegative. Using $|Q| \ge 1 - \bar{A}$ when dividing gives

$$\operatorname{Re}(V/Q) \le \frac{\bar{B} - 2Z}{1 - \bar{A}} - (D_K - \operatorname{Re} T). \tag{9}$$

The terms containing D_K are combined algebraically before enclosure. They do not receive separate upper and lower bounds that would lose the cancellation. Substitution in (5) gives

$$S > F + \frac{D_K - \operatorname{Re} T}{2} + \left(E_\ell - \operatorname{Re}(E_h/E)\right), \qquad F = \ell + \frac{C}{2} - \frac{\bar{B} - 2Z_{\mathrm{low}}}{2(1 - \bar{A})}. \tag{10}$$

The extra overlap is a positive finite sum, so it can be replaced by the certified lower enclosure Z_{low}. This estimate and the cone-surplus estimate bound the same physical quantity. Either may be selected, but their gains cannot be added.

4.3 Paired logarithmic and grouped-head alternatives

At $x \pm a$, the first-head phases satisfy $z_d^{\pm} = z_d^c \exp(\pm i a \log d/2)$. From (6) and Lemma 3, a complete source-class pair $j \le k$ has the following guaranteed logarithmic deficit:

$$G_{k,a} = \sum_{2\le d\le K} \widetilde{d}_{t_-}(d) d^{-\sigma_{k,+}} \inf_{a\in I_a} \left(1 - |\cos(a \log d/2)|\right), \quad \sigma_{k,+} = (1 + h)/2 + t_+ L_{k,+}/4.$$

The higher upper abscissa supplies a coefficient lower bound valid at both points. Therefore

$$\frac{S(x - a, h, t) + S(x + a, h, t)}{2} > \frac{F_j + F_k + G_{k,a}}{2} + G_{jk,a}^{\mathrm{Euler}}. \tag{11}$$

The finite logarithmic estimate is certified on the first 320 time rows and the five bounded frequency classes. The remaining rows and the unbounded class use their complete cone/Euler estimates. The assembler enforces these domains and does not extrapolate beyond them.

A separate grouped-head alternative retains pairs with both indices in $[2, 2048]$, disjoint from the constant/head pairs. Its signed coefficients and each complete group phase interval are retained; the same identity (6) supplies the deficit. It is an alternative to (11). For each class pair, the assembler selects the largest valid lower bound. It then takes the minimum over all eleven class pairs. This selection rule avoids counting any shared contribution twice.

5 Retuned physical source and exact selection

The physical observable remains

$$S(x,h,t) = -\operatorname{Im}\frac{H_t'}{H_t}(x+ih), \qquad \frac{S(x-rh,h,t)+S(x+rh,h,t)}{2} \geq \ell_{i,r}.$$

Probe nonvanishing is established before division, as in the retained source theorem [12]. No old lower bound is interpolated in h. The all-frequency source region, native-cutoff treatment, two reflected branches and corrected Gaussian error argument are unchanged.

5.1 Coverage and production scope

The new frozen plan contains 608 closed time cells $[i/4000, (i+1)/4000]$, $0 \leq i < 608$, through time 0.152. There are six frequency ranges per cell, including an unbounded final range. Every same-class pair and every adjacent-class pair is included: six plus five possibilities. The physical source starts at $X_0 - 2$, where $X_0 = 6{,}000{,}000{,}185{,}827$ and $N_0 = 690{,}988$. All future native cutoffs in the final range remain covered.

Relative to the old plan, exactly 274 heights change. The new heights lie between 0.69 and 1.88; the old maximum was 2.18. For example, the first height becomes 47/25 rather than 109/50. The source producer's enlarged analytic height domain already covers this range. Its Cauchy-disk, activation, mass and denominator acceptance conditions are checked at the actual time–height cells, not on an invalid rectangle formed by pairing the smallest height with time zero.

All 608 cells are freshly produced in both constructions, including the cells whose height did not change. Construction A uses 192-bit interval arithmetic and integer-bin division 800; construction B uses 256 bits and division 900, with a different derivative construction. Each is produced in four adjoining blocks, with endpoints 0–151, 152–303, 304–455 and 456–607. Complete arrays and receipts are reconstructed after every block is present. Both constructions use the same bundled Arb/FLINT library; they are not independent transcendental implementations.

Exact source reconstruction checks 127,680 acceptance inequalities and 26,752 overlaps of the two constructions' temporal interval enclosures. The corrected Gaussian and physical-error safety checks pass on all 1,216 error rows. The 256-bit and 384-bit finite-head wall files are retained byte-for-byte. Those wall computations, and the published finite-RH computation, are not represented as fresh production.

5.2 Matched logarithmic coefficients at the new heights

The joint translated-source producer is rerun in both constructions over the complete schedule. Its exact reassembly checks 1,284,096 conditions. The matched-logarithmic producer is then rerun at the same new heights, with the original finite logarithmic head of 8,192 on the first 320 cells, common native head 690,988, and head-pair groups through 2,048 on all 608 cells [11].

The improvement retains the original order of operations: common finite convolution terms are cancelled before absolute values are taken, and a complete bound is formed only after all remaining errors and the logarithmic-derivative denominator are paid. This yields 9,120 positive head-pair credits and 6,384 improvements from the matched-log alternative in this reassembly. Cross-construction checks agree on 608 heat-coefficient grids, 315 available logarithmic-phase grids, and 560 head-group phase grids. A cache entry without a logarithmic grid is not treated as though it had one.

The focused 16-band source producer is *not* rerun at new heights. Instead, the new valid broad continuum bound is used in every narrow band. At cells with exactly the same height and time as the old source, the maximum of the two complete valid bounds is allowed. This is

a maximum of whole certified estimates, never an unverified mixture of a numerator from one construction and a denominator from another.

5.3 A frozen whole-cell choice

A rational discovery pass selects either the old enhanced source or the new complete source in each cell. The resulting list of 608 provider indices is frozen before the final histories are regenerated. It selects 466 new-source cells and 142 old-source cells; 274 selected heights actually differ from the original heights. A new provider may improve the floor even where its height is unchanged.

For every selected cell, the height and entire row of lower bounds are copied from the chosen provider. The independent reconstruction checks this operation verbatim and binds the resulting source to all four comparison configurations by SHA-256. Neither a sampled optimum nor greedy global optimality is a premise. Any complete, valid frozen choice would give a legitimate comparison; this particular one gives (1).

Proposition 4 (Selected physical source)**.** *For every cell I_i and either consumed shift $r = 23/40$ or $r = 3/5$, the selected source supplies a rational lower bound $\ell_{i,r}$ such that*

$$\frac{S(x - rh_i, h_i, t) + S(x + rh_i, h_i, t)}{2} > \ell_{i,r} \qquad (t \in I_i,\ x \geq X_0).$$

The assertion includes all six frequency classes, all eleven translated class pairs and every future native cutoff in the unbounded class.

Proof. The physical approximation and corrected remainder give (5). The common-phase estimate (6), the matched overlap (8) and the paired alternatives give valid bounds for each class pair. The interval producers enclose their finite masses, derivatives and phases on the complete time–height–frequency cells. The exact reconstruction checks the activation and denominator conditions, selects the strongest complete alternative for each pair, and then takes the minimum over all eleven possibilities. The selected source copies a complete valid row, including its height, from one provider. The reconstruction checks this copy exactly. Thus selection preserves each provider's all-frequency statement. □

6 Full-domain kernel comparison

At a simple highest upper zero $x + iy$, write $q = y^2$. The canonical zero-motion identity, grouped into conjugate pairs, is

$$q' = -2 - 4y \sum_{\text{other pairs}} K_y(d_j, b_j). \tag{12}$$

The own conjugate contributes -2 exactly once. Real zeros have half-pair weight, and the paired imaginary sums converge absolutely [5, Proposition 3.1]. The task is to bound the remaining sum from the source at the higher probes.

For $h > 0$, define the conjugate-pair kernel

$$K_h(d, b) = \frac{h - b}{d^2 + (h - b)^2} + \frac{h + b}{d^2 + (h + b)^2}. \tag{13}$$

A translated observation at displacement rh uses

$$J_{h,r}(d, b) = \frac{K_h(d + rh, b) + K_h(d - rh, b)}{2}.$$

The useful inequality has the form

$$2K_y(d,b) \geq \lambda \frac{y}{h} J_{h,r}(d,b), \qquad 0 < y \leq \rho h, \quad 0 \leq b \leq y, \quad d \in \mathbb{R}. \tag{14}$$

The point $(d,b) = (0,y)$ is excluded because K_y is undefined there. It is the distinguished contact pair, removed before this inequality is summed. A multiple contact is treated by the separate contact argument; no artificial value or path-independent limit is assigned at the singular point.

Proposition 5 (New certificate family)**.** *All 44 triples (r,ρ,λ) in Appendix A satisfy* (14)*. The shifts are $r = 23/40$ and $r = 3/5$. For each shift, the 22 values of 100ρ are*

$$5, 10, 15, 20, 25, 28, 30, 31, 32, 33, 34,$$
$$35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45.$$

Every λ is an explicitly stored positive rational, at most two. For example, the first two actively used domain refinements include

$$(r,\rho,\lambda) = \left(\frac{23}{40}, \frac{2}{5}, \frac{99641}{50000}\right), \qquad \left(\frac{3}{5}, \frac{2}{5}, \frac{198313}{100000}\right).$$

Proof. Scale h to one and put

$$D = d^2/h^2, \qquad v = y^2/h^2, \qquad B = b^2/h^2, \qquad R = r^2.$$

Introduce the polynomials

$$\begin{aligned} C &= D + R + 1 + B, & A &= C^2 + 4DR - 4B, \\ D_4 &= A^2 - 16DRC^2, & G &= A(C - 2B) - 8DRC, \\ Q &= (D + v - B)^2 + 4DB. \end{aligned}$$

Pairing the two translations first removes the square root of D. After multiplication by positive denominators, (14) is equivalent to

$$P = (D + v - B)D_4 - \frac{\lambda}{2} QG \geq 0. \tag{15}$$

The translated denominators are positive because $b/h \leq \rho < 1$. The remaining denominator vanishes only at the excluded point.

Set $D = z/(1-z)$, $v = \rho^2 p$, $B = \rho^2 pq$. Then

$$\widetilde{P}(z,p,q) = (1-z)^5 P\big(z/(1-z), \rho^2 p, \rho^2 pq\big)$$

is a polynomial of degree at most five in each variable. The required finite domain is the cube $0 \leq z,p,q \leq 1$, with $z = 1$ included by polynomial continuation. Each certificate stores its rational coefficients, up to an independently checked positive scalar, and a complete binary subdivision of this cube. On every leaf, all tensor Bernstein coefficients are nonnegative. Since the Bernstein basis is nonnegative on the cube, this proves the assertion.

The independent verifier does not import the searcher's polynomial algebra. It evaluates the closed expression (15) and the stored polynomial on the exact tensor grid $\{0, 1/6, \ldots, 5/6\}^3$. The degree bound and six distinct nodes per variable make this an unisolvent polynomial-identity check, not sampled positivity. It also checks the positivity of the scalar normalisation. For a degree-n Bernstein coefficient vector b, the two half-interval restrictions are independently computed by

$$b_k^L = 2^{-k} \sum_{j=0}^{k} \binom{k}{j} b_j, \qquad b_k^R = 2^{-(n-k)} \sum_{j=k}^{n} \binom{n-k}{j-k} b_j.$$

These matrices are applied along the chosen coordinate at each split. The partition verifier requires both children, forbids duplicate leaves and forbids a leaf together with one of its descendants. Thus there is no unverified gap in the cube cover. The complete run checks 9,504 exact identities, 331 leaves and 71,496 Bernstein coefficients. Seven deliberately corrupted inputs are rejected. □

6.1 The whole distinguished-pair subtraction

At a zero of height y, write $w = y^2/h^2$. The distinguished pair's normalised contribution at the translated probes is

$$s_r(w) = \frac{2(1 + r^2 - w)}{(1 + r^2 - w)^2 + 4r^2 w}. \tag{16}$$

The constants used by the new certificates are

$$M_{23/40} = \frac{1503067131}{10^9}, \qquad M_{3/5} = \frac{25}{17}.$$

The bound $s_r(w) \le M$ for $0 \le w \le \rho^2$ is verified by checking nonnegativity of the quadratic

$$Mw^2 + \{M(2r^2 - 2) + 2\}w + M(1 + r^2)^2 - 2(1 + r^2). \tag{17}$$

Its endpoints and any interior minimum are checked exactly. The scalar auditor separately checks the same condition on the full incoming comparison interval at every step. An own-pair cap at just one sampled height would not suffice.

If the physical translated source has lower bound ℓ, summing (14) after removing the distinguished pair gives the rate

$$q' < -2 - cq, \qquad c = \frac{2\lambda}{h}\left(\ell - \frac{M}{h}\right) > 0, \qquad q = y^2. \tag{18}$$

The -2 is the own-conjugate contraction. The M/h term subtracts the entire contribution of that pair from the source; it is not optional. Real zeros retain the half-weight convention of the zero-motion formula.

The history uses only fixed shifts. The retained code also supports a continuum sharp mode, but no main-history step selects it. In that mode one must evaluate the displacement at the actual contact height and cover every swept displacement band; freezing the incoming displacement with the sharp incoming self cap is invalid. The source supplement records that correction explicitly. The present fixed-shift proof instead uses (17) on the entire incoming height interval.

7 Global zero-height comparison

Every time cell is divided into 32 equal pieces:

$$\Delta t = \frac{1}{128000}.$$

At each incoming certified upper value q, choose an admissible positive rate (18). Its comparison solution is

$$Q(t + \tau) = \left(q + \frac{2}{c}\right) e^{-c\tau} - \frac{2}{c}. \tag{19}$$

The program rounds each endpoint upwards and uses that upper bound to start the next step. Another strip estimate or spatial guard never lowers the restart value.

For a step starting at q, the selected certificate must satisfy $q/h_i^2 \le \rho^2$ and $|rh_i| < 2$. The comparison decreases, so the same kernel domain holds throughout the step. The own-pair polynomial is checked on all $0 \le w \le q/h_i^2$. These whole-interval tests are what allow a fixed translation to be used safely; a self cap at just the incoming height would not be enough.

Lemma 6 (Global first-contact comparison)**.** *Under the hypotheses and certificates above, the piecewise flow* (19) *and its upward restarts enclose the squared imaginary height of every zero through the declared terminal step.*

Proof. The first-rate check gives

$$q_0 - (2 + c_0 q_0)10^{-8} > 9/16.$$

The comparison is convex and above its tangent. Together with de Bruijn's contraction from the initial strip, this establishes a positive interval of strict separation.

On a closed time interval bounded away from zero, positive-time confinement in [5, Theorem 1.5(i)] confines possible nonreal zeros to a bounded real range. The wall localises a contact to $|x| > X_0$. Thus a first-contact argument takes place on an actual compact zero set; no possibly unattained global supremum is differentiated.

A multiple upper contact is impossible: the backward Hermite splitting of [5, Proposition 3.1(ii)] produces an upper branch at an earlier time with height excess of order $\sqrt{\epsilon}$, whereas the positive barrier changes by $O(\epsilon)$. This contradicts previous separation. At a simple contact all other zero heights are below the barrier, so the full kernel inequality and source floor apply. The contact derivative is strictly smaller than the comparison derivative by (18), excluding a first crossing. Equivalently one can use the upper Dini derivative of the finitely many local simple branches and integrating-factor comparison. At a time knot the preceding left-hand inequality applies; an upward restart cannot create a contact. These alternatives cover every prospective first contact. □

Stop the final step at $q_e = 10^{-8}$. The exact comparison travel time is

$$\tau_e = \frac{1}{c} \log\left(1 + \frac{c(q - q_e)}{2 + cq_e}\right). \tag{20}$$

The auditor verifies $0 < \tau_e \le \Delta t$, ensuring that the last source bound remains valid throughout the terminal interval. Lemma 6 and ordinary strip contraction give

$$\Lambda \le t + \tau_e + \frac{q_e}{2}. \tag{21}$$

The bound includes the full final strip-contraction time $q_e/2$. If all zeros become real sooner, the same upper bound still holds.

8 Exact endpoint and reproducibility

8.1 Two rational constructions

The primary producer bounds e^{-x}, $0 \le x < 1$, by its even degree-18 Taylor sum and rounds squared-height endpoints upward on the 2^{-96} grid. The alternate uses the reciprocal of the positive degree-12 Taylor sum for e^x and the 2^{-112} grid. For the final $\log(1 + z)$, $0 < z < 1$, the primary uses the odd degree-11 sum and the alternate uses $z - z^2/(2(1 + z))$. Both total terminal times are rounded upward to multiples of 10^{-15}.

The separate scalar auditor imports neither the producer nor its selector. It uses an even degree-20 exponential sum and an odd degree-13 logarithmic sum with exact powers and factorials. It verifies the hypothesis and seed, initial delay, source identity, every time join, kernel eligibility, complete own-pair subtraction, every positive rate, all outward endpoints and the terminal contraction. Agreement with the producer is required at the exact inequality level, not just in displayed decimal digits.

Construction	Steps	Exact decimal ceiling
Primary	14,899	0.116397726408291
Alternate	14,899	0.116397726408312

Each construction uses $r = 23/40$ on 1,024 steps and $r = 3/5$ on 13,875 steps. The terminal source index is 465, and the final substep starts at 7449/64000. Its permitted source interval ends at 14899/128000. The exact final logarithmic travel time lies inside that interval. Both histories have been regenerated byte-for-byte from the selected source and checked by the separate auditor. Corrupted-history rejection tests are retained with the verification receipts.

Proof of Theorem 1. Lemma 2 gives the initial strip and strict separation. The corrected wall localises every possible nonreal contact. Proposition 4 and the full-domain kernel certificates supply the strict rate bound (18) at every simple contact; Lemma 6 handles the global comparison, including multiple contacts and time joins. The independently audited primary history bounds the terminal travel time in (20). Adding the complete $q_e/2$ contraction allowance and rounding upward gives (1). □

8.2 Separating the verification layers

A hash identifies data; it does not establish a physical inequality. The evidence therefore separates the analytic derivation, interval production, exact reconstruction and scalar audit.

Obligation	Written argument	Executable check
Initial strip	Section 2, conditional on $\mathsf{H}_{7/8}$	Seed and positive-delay tests
Physical source	Sections 3–5; source supplement; Foundations	Raw interval records, coverage and complete exact reconstruction
Gaussian error and wall	Appendix B; Foundations	Safety inequalities and retained wall certificates
Kernel and self pair	Section 6; Appendix A	Independent polynomial identity, full cube cover and self-cap tests
Global comparison	Section 7	Complete joins and source binding
Terminal bound	This section	Two rational histories and separate auditor

`Proof_and_Replay.zip` contains the frozen scientific evidence, raw source enclosures, original production receipts, analytic source texts and all eight final-source histories. Only the two configurations labelled `main` prove the headline under $\mathsf{H}_{7/8}$; the other six use separately stated initial strips. The *Verification Guide* gives the integrity-only, stored-source and native-production commands. The stored-source audit reconstructs the full source and reruns the exact comparisons. The optional native replay reruns the transcendental producers at the frozen retuned plan. These are different operations.

The two native constructions use the same Arb/FLINT library [10], with different precision, integer partition and derivative construction. The finite-head wall data are retained, not regenerated in the retuning; the published finite-RH computation is not repeated. The programs do not prove the external 7/8 assertion or formalise the complete analytic argument. No external referee certification, unconditional record, global optimum or proof of RH is claimed.

Release and attribution. This is version 1.0 of the first deposit of the 0.116398 conditional paper. Earlier 0.118 files remain in the immutable evidence as *undeposited development drafts*. The separate September record [1] retains its own identity and review status.

AI assistance. The AI assisted with the derivations, code, numerical work and exposition. The named author is responsible for the submitted claims. The supplied checks and explicit assumptions, rather than this disclosure, define the verification scope.

A Exact kernel constants

The table lists every new kernel certificate. Each displayed decimal is exact. Each row has a complete rational polynomial and subdivision certificate in `evidence/GEOMETRY_CERTIFICATES.json`. The admissible domain and its independent verification are proved in Section 6. The whole-interval self caps are $1503067131/10^9$ for $r = 23/40$ and $25/17$ for $r = 3/5$.

ρ	λ for $r = 23/40$	λ for $r = 3/5$
0.05	1.99998	1.99819
0.1	1.99997	1.99784
0.15	1.99992	1.99719
0.2	1.99968	1.99612
0.25	1.99911	1.99448
0.28	1.99853	1.99312
0.3	1.998	1.99202
0.31	1.99768	1.9914
0.32	1.99733	1.99074
0.33	1.99694	1.99002
0.34	1.99651	1.98925
0.35	1.99603	1.98841
0.36	1.99551	1.98751
0.37	1.99493	1.98654
0.38	1.99429	1.98549
0.39	1.99359	1.98436
0.4	1.99282	1.98313
0.41	1.99197	1.98179
0.42	1.99105	1.98033
0.43	1.99003	1.97036
0.44	1.96906	1.93243
0.45	1.92508	1.89014

B Gaussian remainder correction and a safe wall

This appendix replaces the 5/6 Gaussian majorant in the retained Foundations, Part A, Section 4.1, and the narrower historical `physical_eC0` wall comparison. The original archival source is preserved. The broad physical error allowances used by both the original and retuned source constructions remain valid.

Let $T' = T + \pi t/8 \geq T$ and $D = T - 3$. In the all-real-abscissa Stirling/Taylor comparison accompanying [5, (62)], the valid coarse inequality is

$$\frac{1}{12(T'/2 - 0.33)} \leq \frac{1}{6(T-3)},$$

not the expression with numerator $1/3$ over $4(T-3)$. With the corrected numerator $2/3$,

$$\frac{u^2 + 6|u| + 2/3}{4D} \leq \frac{u^2 + 11/12}{D},$$

because multiplication of the difference by $4D$ gives $3(|u|-1)^2$. The Gaussian integral is therefore

$$\begin{aligned}\widetilde{G}_\sigma(c) &= \int_{\mathbb{R}} e^{(u^2+11/12)/D+cu}\frac{e^{-(u-\sigma)^2/t}}{\sqrt{\pi t}}\,\mathrm{d}u \\ &= \frac{\exp\left\{\dfrac{11}{12D}+\dfrac{\sigma^2/D+\sigma c+tc^2/4}{1-t/D}\right\}}{\sqrt{1-t/D}}.\end{aligned} \tag{22}$$

At $t=0$ use its limiting value. Every actual disk satisfies $-1\le\sigma\le 2$, $0\le t\le 1/5$ and

$$D\ge (X_0-4)/2-3>10^{12}.$$

The elementary inequality $-\log(1-z)\le z/(1-z)$ gives

$$\log\widetilde{G}_\sigma(0)<6/D,\quad \widetilde{G}_\sigma(0)-1<10^{-10}.$$

For $|c|\le 11/5$ the same explicit formula gives $\log\widetilde{G}_\sigma(c)<93/20$ and $\widetilde{G}_\sigma(c)<110$. This covers the required slopes $\log 9$ and $-\log 2$. The numerical constant inequalities are enclosed by rational positive Taylor sums and geometric tails.

The two all-real-σ coefficient errors in [5, (62)–(64)] therefore satisfy

$$\delta_1+\delta_2\le\frac{(0.2+0.029)110}{N_0-1}<10^{-3}.$$

For the far negative Gaussian tail $u=-r$, $r\ge 100$, the complete coefficient majorant in the Foundations bounds the integrand by

$$\exp\{-4r^2+20r+13+(r^2+1)/D\}\le e^{-3r^2}.$$

The quadratic difference is positive and increasing from $r=100$ for $D\ge 100$, and $\int_{100}^{\infty}e^{-3r^2}\mathrm{d}r\le e^{-30000}/600<10^{-100}$. Thus the full remainder bracket is bounded by

$$\tfrac{1}{2}(1+10^{-10})+\frac{(0.2+0.029)110}{N_0-1}+10^{-100}<1.$$

This establishes the retained coarse remainder bound:

$$|R_{t,N}(\sigma+iT)|<e^{t\pi^2/64}|M_0(iT')|.$$

Comparison with the physical normaliser then gives the same broad physical error bound

$$e_C^{\mathrm{broad}}<4\exp\left\{-\frac{(1+y)L}{4}-\frac{tL^2}{16}+\frac{1}{10}\right\}. \tag{23}$$

The existing producers subtract the allowance (23) on their actual Cauchy disks, together with the complete edge pair and the leading-model value and derivative error bounds. The separate safety checker verifies the 29 rational conditions and 1,216 physical-error rows for each complete two-construction source. It is applied separately to the original and the retuned data.

At the wall (23) is largest at $t=y=0$. Using $\pi<22/7$, $e^{1/10}<10/9$ and $7X_0/88>820^4$ gives

$$e_C^{\mathrm{broad}}<\frac{40}{9\cdot 820}=\frac{2}{369},\qquad \frac{2}{369}+10^{-6}<\frac{1}{180}.$$

The two wall certificates give ideal-model floor $79/100$, full ideal displacement below $1/100$, and $e_A+e_B<10^{-6}$. Hence

$$|H_t/B_t|>79/100-1/100-1/180=697/900>3/4.$$

The narrower historical wall-error field is not used as a premise. The archived documents and hashes remain unchanged, and the new receipt records the replacement inequalities explicitly.

References

[1] D. J. Raymond, *Correlated translated probes and the bound* $\Lambda < 0.149396$ *for the de Bruijn–Newman constant*, Zenodo preprint, 21 September 2026, doi:10.5281/zenodo.22867547. Includes *Analytic Foundations* and `Proof_and_Replay.zip`. Public companion revision `3e519044ed8f2f969d76dc0c34008727302e67ec`.

[2] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\mathrm{Re}(s) > 7/8$, OpenAI Math Release preprint `OAI:The-Quasi-Riemann-Hypothesis-September-30-2026`, 2026. Pinned manuscript directory.

[3] OpenAI, *The Quasi-Riemann Hypothesis*, 5 October 2026. Pinned 11/12 companion.

[4] OpenAI, `OAI.riemannZeta_ne_zero_of_seven_eighths_lt_re`, `Nonvanishing.lean`, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, Pinned supplied source identity. No Lean execution is reported here.

[5] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann* ξ *function, and a new upper bound for the de Bruijn–Newman constant*, Research in the Mathematical Sciences **6** (2019), article 31. doi:10.1007/s40687-019-0193-1. Numbered references follow arXiv:1904.12438v2.

[6] D. J. Platt and T. S. Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, Bulletin of the London Mathematical Society **53** (2021), 792–797. doi:10.1112/blms.12460; arXiv:2004.09765.

[7] J. Gomila, *A computer-assisted proof of the bound* $\Lambda \leq 893927/5000000 = 0.1787854$ *for the de Bruijn–Newman constant*, 2026 manuscript cited in the retained earlier source. Pinned source cited by the prior publication; context only.

[8] S. Gordon, *A computer-assisted upper bound for the de Bruijn–Newman constant*, 2026 manuscript cited in the retained earlier source. Pinned source cited by the prior publication; context only.

[9] M. Planat, *Collective Contraction in the de Bruijn–Newman Heat Flow: Off-Zero Logarithmic Derivatives and Certified Barrier Refinements*, 2026, arXiv:2609.37164.

[10] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Transactions on Computers **66** (2017), 1281–1292. doi:10.1109/TC.2017.2690633.

[11] D. J. Raymond, *Matched logarithmic source: analytic supplement*, version 1.0, 10 October 2026. Included as `Matched_Source_Supplement.pdf`; original analytic source and producer code retained in the accompanying `Proof_and_Replay.zip`.

[12] D. J. Raymond, *Coupled probe motion source*, `BCLD_COUPLED_PROBE_MOTION_SOURCE_20260921`, `PROOF.md`, Sections 4–12. Original analytic text and producers retained in the evidence dependency archive; typeset in *Prior Analytic Foundations*. The Gaussian replacement is Appendix B of the present paper.